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Use of Artificial Intelligence tools in supporting decision-making in hospital management

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Abstract

Background The use of Artificial Intelligence (AI) tools in hospital management holds potential for enhancing decision-making processes. This study investigates the current state of decision-making in hospital management, explores the potential benefits of AI integration, and examines hospital managers' perceptions of AI as a decision-support tool.

Methods A descriptive and exploratory study was conducted using a qualitative approach. Data were collected through semi-structured interviews with 15 hospital managers from various departments and institutions. The interviews were transcribed, anonymized, and analyzed using thematic coding to identify key themes and patterns in the responses.

Results Hospital managers highlighted the current inefficiencies in decision-making processes, often characterized by poor communication, isolated decision-making, and limited data access. The use of traditional tools like spreadsheet applications and business intelligence systems remains prevalent, but there is a clear need for more advanced, integrated solutions. Managers expressed both optimism and skepticism about AI, acknowledging its potential to improve efficiency and decision-making while raising concerns about data privacy, ethical issues, and the loss of human empathy. The study identified key challenges, including the variability in technical skills, data fragmentation, and resistance to change. Managers emphasized the importance of robust data infrastructure and adequate training to ensure successful AI integration.

Conclusions The study reveals a complex landscape where the potential benefits of AI in hospital management are balanced with significant challenges and concerns. Effective integration of AI requires addressing technical, ethical, and cultural issues, with a focus on maintaining human elements in decision-making. AI is seen as a powerful tool to support, not replace, human judgment in hospital management, promising improvements in efficiency, data accessibility, and analytical capacity. Preparing healthcare institutions with the necessary infrastructure and providing specialized training for managers are crucial for maximizing the benefits of AI while mitigating associated risks.

Keywords Artificial Intelligence, Decision making, Digital transformation, Hospital management

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Background

The implementation of computer systems and digital tools has proven crucial in improving healthcare and the safety of patients and users of healthcare services. The cross-sectional use of clinical technology highlights significant advances in professional documentation standards, data access, and platforms for handling clinical, health, and administrative information, indicating a movement toward digitalization as a fundamental strategy to face the current challenges of the sector [1–4].

In the current context of increasing demand and competitiveness in the healthcare market, streamlining procedures and processes emerges as imperative to providing healthcare services that add tangible value to both users and society. This need is accompanied by an increase in the complexity and volume of necessary responses, imposing high standards of management and clinical quality, focusing on efficient resource management, equity of access, safety, risk management, and sustainability [5–7].

The provision of excellent services and the definition of processes consistent with the population's needs guarantee access and quality of care and must be continuously considered. Thus, the existence of efficient management mechanisms, agile processes, and objectivity in the professional performance of managers is crucial for equity and social justice in the right to health, optimization of resource use, and quality organizational performance [8, 9].

The opportunity for digital transformation in healthcare organizations and professionals' performance is not limited to the clinical aspect, requiring a broader vision that includes hospital management [10]. Health institutions have been experiencing this path of change, with the development of a diverse set of digital platforms and tools supporting the performance of middle managers, including hospital administrators and health professionals with management roles, as well as supporting decision-making for top managers in local, regional, or national organizations.

However, hospital management currently faces numerous challenges in health systems, such as decision complexity, information variability, data volume/big data, timing impositions, intervention intensity, limited interoperability, low data integration, and differing digital competencies among professionals. Consequently, existing digital systems do not yet seem to be facilitating elements [6, 11].

The continuous development of technology necessarily opens new opportunities for reflection and enhances the evolution of what is currently available to the various agents responsible for health management. The evolution of information systems, especially with the development of increasingly complex and advanced models of artificial intelligence (AI) and machine learning (ML), and their application in clinical tools, seems increasingly inevitable, as evidenced in different fields and areas of society [3, 12].

Integrating AI into clinical systems faces risks such as a lack of transparency or excessive dependence. On the other hand, challenges include slow adoption, mistrust, undervaluation, and the need for rigorous consideration to avoid violations of patients' rights or compromising the quality of care provided [13–17].

The diversity of treatments and therapeutic processes, along with the constant evolution of evidence, results in a complexity of variables and data volume that exceed human capacity for management and processing, highlighting the need for systems that can effectively assist in decision-making processes [12, 14, 18].

There are AI applications that promote resource optimization and service efficiency through the analysis of patients' data history, bed occupancy, and human resources [19, 20]. Population risk stratification, personalized patient education, and simplification in scheduling consultations and exams are other examples of its importance in providing healthcare [21–23]. This type of technology allows for analytical, predictive, and even strategic capabilities. Thus, recognizing its potential will dictate a path of change in the pattern of digital platforms and a significant window of evolution in conducting analysis, evaluation, and decision-making interventions in healthcare organizations [2, 24–27].

Financially, the absence of AI-driven decision-making tools in healthcare facilities can lead to significant inefficiencies in resource allocation, resulting in increased operational costs. For instance, a study published in *Health Affairs* estimated that inefficiencies in the U.S. healthcare system could amount to as much as $935 billion annually, largely due to administrative complexities and unnecessary services [28]. These challenges underscore the potential impact of AI tools, which can streamline decision-making processes, optimize resource use, and reduce operational costs [29]. Supporting this potential, a study by Accenture projects that AI applications in healthcare could generate up to $150 billion in annual savings in the U.S. by 2026 [30].

Therefore, this study aims to address the specific needs of hospital management by exploring the current use and challenges of AI tools in supporting decision-making. The following specific objetives were defined:

1. To analyze existing decision-making practices to identify areas for improvement and opportunities for integrating AI;
2. To evaluate the potential benefits of AI in optimizing resources and improving decision quality in healthcare management;

3. To investigate the perceptions of hospital managers regarding AI adoption, including barriers and facilitators to its implementation;
4. To identify key challenges and determinants for the effective integration of AI tools in hospital management decision-making processes;
5. To provide a focused and structured exploration of how AI can address the complexities and demands of modern hospital management, ensuring a systematic approach to understanding the integration of AI in healthcare decision-making.

Methods

Considering the objectives intended with this study, a descriptive and exploratory investigation was applied, using a qualitative approach aimed at understanding and characterizing the phenomenon under study [31]. The purpose of data collection was to capture a broad range of experiences and opinions about the challenges, opportunities, and implications of integrating AI tools and decision support into management in healthcare organizational contexts, thus contributing to a richer and more multifaceted understanding of the topic.

Data collection happened between March and April 2024 and was carried out through semi-structured interviews designed specifically for this study. This approach was chosen to capture the unique perceptions of hospital managers regarding existing decision-support tools, and the functionalities and capabilities of AI tools in supporting decision-making in hospital management. The target population includes middle and senior managers, hospital administrators, and clinicians with management roles, all of whom are integral to decision-making processes that require consulting, analyzing, evaluating, documenting, predicting, and monitoring data from various tools and digital platforms at both institutional and national levels.

The recruitment of these participants ensured a balanced and diverse representation of different departments, hierarchical levels, and areas of specialization and expertise from various institutions (Table 1). This multidisciplinary approach allowed for a more comprehensive and diverse view of the use and impact of AI in hospital management.

An interview guide was developed, as existing tools were found inadequate to fully address this study's specific research questions and objectives. The questions were supported by relevant literature and aligned with the study's objectives (Additional file 1).

A pre-test of the interview script was conducted to refine the questions and prepare the interviewer/researcher. Prior to the start of each interview, time was dedicated to presenting the researcher and explaining the study objectives, allowing participants to become familiar with the topic under investigation. This preparation ensured that the interviewees were comfortable and engaged, facilitating a deeper and more meaningful data collection process.

The ethical integrity of the study was duly ensured with the prior obtaining of informed consent from all participants, thereby guaranteeing their informed knowledge about the purpose of the research, treatment of the collected data, and also their right to withdraw and exit the study freely and at any point during the investigation. The study was approved by the Ethics Committee of NOVA-NSPH.

The interviews were audio-recorded and transcribed verbatim. The data were anonymized at the time of recording and treated with utmost rigor and respect for the privacy of those involved, in line with the ethical principles of scientific research, and at no point was the identity of the interviewees shared. Thus, after conducting the interviews, textual transcription was carried out with accuracy and respect for the transmission of information from the interviewees, properly identified with an alphanumeric designation - for example, "E1." Access to the recordings and all resulting data remained restricted to those involved in the research and was secured through a coded digital folder.

Following the transcription of the interviews, the data were coded manually using a deductive approach guided by established theoretical concepts. This process involved identifying and categorizing themes, patterns, and emerging ideas from the participants' responses based on a predefined set of topics, ensuring a systematic and focused methodological approach. The data structuring was grounded in predefined variables, allowing for broad coding that identified general themes as well as more specific associations. This deductive approach facilitated a detailed and structured analysis of how theoretical frameworks manifested within the specific context of the research, enabling a comparison between theoretical expectations and the real-world experiences and perspectives of participants. Additionally, the manual coding process was chosen for its flexibility in accommodating nuanced interpretations of the data, enhancing the depth of the analysis.

This comprehensive approach aimed not only to analyze the results in light of existing knowledge but also to provide a deeper understanding of the phenomena under investigation, thereby ensuring the rigor and transparency of the qualitative analysis process.

The number of interviewees ($n=15$) took into account the verification of sample saturation, that is, until the same ideas and concepts recurred in the participants' responses.

Table 1 Overview of interviewees

Interviewer	Qualifications	Professional Category	Position(s)	Experience
E1	Degree in Economics PgD in Hospital Administration	Hospital Administrator	Hospital Department Management (Board Member) Integrated Responsibility Center (Board Member)	14 years
E2	MSc in Health Management and Economics	Executive Manager (healthcare)	Hospital Department Management (Board Member)	30 years
E3	MSc in Management PgD in Hospital Administration	Hospital Administrator	Hospital Department Management (Board Member) Integrated Responsibility Center (Board Member)	8 years
E4	MBA in Administration PgD in Hospital Administration	Hospital Administrator	Board of Director's	21 years
E5	PgD in Hospital Administration	Hospital Administrator	Department Management (Head)	22 years
E6	PgD in Management PgD in Hospital Administration	Hospital Administrator	Department Management (Board Member)	3 years
E7	MSc in Management PgD in Hospital Administration	Hospital Administrator	Hospital Department Management (Board Member) Department Management (Head)	14 years
E8	MSc in Management PgD in Hospital Administration	Hospital Administrator	Department Management (Head)	6 years
E9	Degree in Economics PgD in Hospital Administration	Hospital Administrator	Department Management (Head) Hospital Department Management (Board Member)	15 years
E10	MSc in Health Management and Economics PgD in Hospital Administration	Executive Manager (healthcare)	Healthcare Manager Hospital Department Management (Board Member)	14 years
E11	Degree in Economics PgD in Hospital Administration	Hospital Administrator	Board of Director's	11 years
E12	MSc in Management PgD in Hospital Administration	Hospital Administrator	Hospital Department Management (Board Member) Integrated Responsibility Center (Board Member)	15 years
E13	MSc in Health Management and Economics PgD in Hospital Administration	Hospital Administrator	Hospital Department Management (Board Member)	18 years
E14	MSc in Management	Manager (healthcare)	Integrated Responsibility Center (Board Member)	9 years
E15	MSc in Management PgD in Hospital Administration	Hospital Administrator	Hospital Department Management (Board Member)	8 years

Results

The analysis of the interviews determined the need for an organized structure for presenting the results in points and subpoints, as shown in Fig. 1. This organization aims to highlight the depth of the hospital managers' responses and support the reader's reflection process.

Current state of decision-making in health management

The interviews with hospital managers reveal a multifaceted reality and a complex landscape of the decision-making process. The content analysis shows that the efficiency of the decision process is influenced by a wide range of factors, from the existence and choice of technological/digital tools, the integration of new management practices and methodologies, the skills for data handling and analysis, and the sensitivity to human aspects.

Isolation in the process and poor communication

The isolated nature of decision-making, often exacerbated by insufficient or difficult access to necessary data: "*There is no practice of exchanging opinions or experiences, and sometimes, the information that (the manager) needs is exactly what is missing*" (E2). "*It is an extremely heterogeneous process*" (E6) "*lonely and often lacking information for a well-founded and safe decision-making*" (E2): reflects a significant issue of internal communication and collaboration among managers and other departments. "*The manager should, effectively, exist to make decisions and should not waste any time handling data"* (E6) and "*when we get to the information, the opportunity has already passed, so the time for intervention has already gone by*" (E12), which suggests the need for a more integrated strategy that promotes the exchange of information and experiences.

Humanization of decision-making

The importance of considering human and subjective elements in decisions is a vital theme. One participant emphatizes, "*it's trying to also listen to the people who may be involved...*" (E9), while another notes, "*the decision process usually starts with an attempt to collect data... but in principle always ends with some subjectivity of analysis and the personal perception of the manager...*" (E15). Despite the emphasis on objective data, individual perceptions and personal experiences still play a crucial role: "*the constrains, more than technological (...), are essentially human*" (E11). "*The main issue is people and how we try to make people adapt to technology, and often we cannot think in a way that allows technology to adapt to people.*" (E11): this statement highlights the complexity of health management, where decision quality relies not only on quantitative analyses but also on the ability to interpret and integrate human and contextual insights.

Inefficiency in data utilization

The use of tools such as Excel® and Business Intelligence systems is a constant in the described management practices. "*We conduct feasibility studies, analyze production data...*" (E1) and "*Other existing tools include some developed dashboards*" (E3) illustrate a blend of traditional and modern methods. This mix suggests a gradual transition to more advanced technologies for presentation and access to information, still limited by the high prevalence of conventional practices – "*I use Excel®a lot*" (E1), a tool that participants unanimously reveal they use. Many managers express frustration with the use and analysis of data, highlighting an urgent need for automation and better integration. The common sentiment is, "*There's an extraction from databases and then work on them*" (E13) in Excel® and "*I already have my maps parameterized... But the truth is if this information was already available... I wouldn't have to spend so much time...*" (E1). "*The efficiency is zero... it's all very manual*" (E7). These quotes underscore the necessity for systems that automate repetitive tasks and integrate data in a useful and cohesive manner.

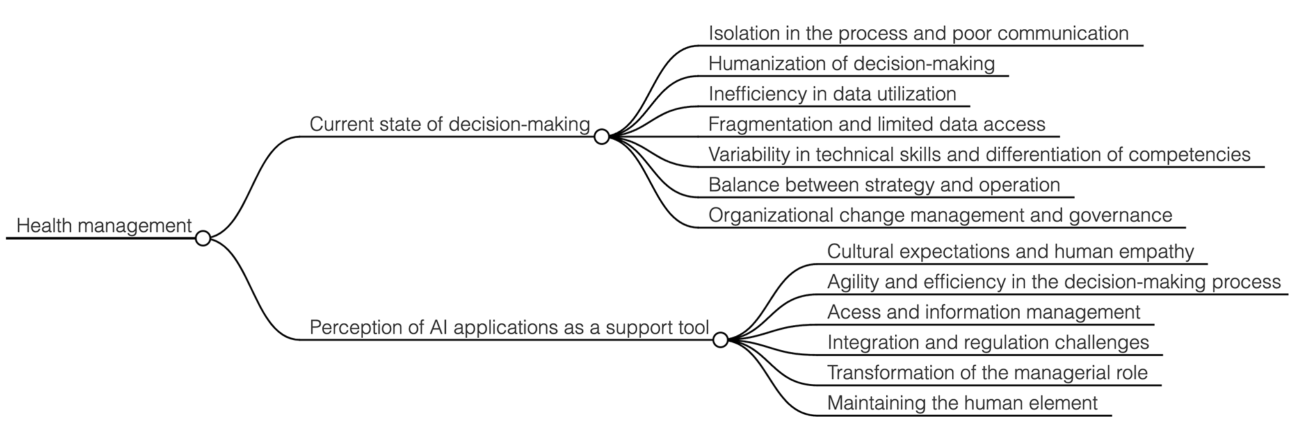

Fig. 1 Mind map of coded interview results with hospital managers

Fragmentation and limited data access

The heterogeneity of data platforms and inefficiencies in decision processes are points of concern. Examples like: "*we have more than 70 web-based applications*" (E4), "*in a large institution perhaps it's not easy to change everything overnight.*" (E6) and "*It should be a process based on indicators, on official data from the service and direct observation*" (E10) demonstrate variations in practices and highlight the need to adopt more effective technologies and more agile and automated management processes. These improvements can positively transform how decisions are made and implemented.

Fragmented information systems (lacking data interoperability) and limited access are often cited as major hurdles. Statements such as "*Often I have to go to two or three systems to be able to gather data*" (E6) and "*The limitation we have is data availability*" (E9), clearly demonstrate the need to consolidate platforms and improve IT infrastructure. Across the board, it is assumed that the efficiency of the decision-making process is "*reduced*" (E1, E3, E6, E7, E10, E12, E14) to "*moderate*" (E2, E4, E8, E9, E15).

The lack of data organization is widely criticized: "*Scattered and nonexistent information with a lack of specification*" (E2) and the need to "*improve the fine specification of information, allocate everything to one application*" (E2) are pointed out as major limitations. These points highlight the urgency to reformulate data management and structures to facilitate access and analysis.

Variability in technical skills and differentiation of competencies

The disparity in technical skills among managers is a recurring issue. As one participant notes, "*Excel® is already a complex tool and not everyone can work with it. In fact, there are people... who don't even know where to go to get the right information*" (E1), complemented by another remark: "*not everyone in hospital administration knows, masters, Excel®*" (E11). These difficulties highlight the crucial importance of personalized training programs that balance and develop the skills of human resources.

Additionally, the shortage of data specialists is another serious issue mentioned by multiple managers. "*We don't have... the capacity to hire people who are experts in data handling*" (E4), and "*We don't have people... we lack people with knowledge about data*" (E5) illustrate this significant gap in human capital, which hinders robust and multivariable data usage, exacerbating the difficulties of the process. In this context, several participants suggest the importance of recruitment strategies for other "new professions" (E15) in the field of data science, while simultaneously advocating for investment in the developing of professionals skills.

Balance between strategy and operation

Managers face the challenge of navigating between daily operational needs and strategic goals. For example, one manager notes: "*The decision-making process always has a more strategic framework... How does the measure we take impact the care we provide to patients?*" (E4). Another highlights the operational aspect, "*At the moment I can tell you that everything goes through the departments... But we end up having some decision-making power here more operationally*" (E7). This fluctuation requires a balance to avoid discontinuities in quality and effectiveness of operations, and to ensure consistency in administration policies and practices: "*this has a lot to do with the leadership and the methodology they bring (the board). Building decision is not easy because hiearchies are not always respected*" (E7).

Organizational change management and governance

Managing organizational change is a particularly challenging area, with managers highlighting cultural resistance: "*It seems to me that the main challenge and also the great benefit are the people*" (E3), and "*the little efficiency... mainly due to the people... to the existing institutional culture*" (E6). Moreover, the participants emphasize the need for strong leadership and inclusive, participatory organizational change programs that promote robust digital transformation strategies, technological adaptation, and innovation: "*The administrations do not listen to the managers*" (E2), "*not only do they not inquire about the adequacy of the available information, but they also do not listen when managers complain about its quality*" (E2).

The lack of clarity in operational rules complicates decision-making, as observed in various statements: "*Sometimes it's difficult because they (the rules) often aren't clear*" (E8) and the "*uncertainty in the outcomes of human decisions*" (E8) is mentioned. Standardization of policies and the implementation of clear guidelines are perceived as essential for improving the efficiency of the process and the security of the decisions: "*We need a digital transformation with someone who knows what AI can do (...), and how we can transpose that to human resources, to scheduling, and to everything that is the support structure of the organization, how all of that can be automated.*" (E15) and "*The efficiency is currently quite fragmented by the need to search for data*" (E10).

Autonomy in data management and greater interoperability between systems and platforms are seen as crucial to overcoming these limitations: "*Some of these additional tools (...) were developed many years ago and provided some competitive advantage, but in any case, generally, they are still not for supporting decision-making.*" (E3). "*The information system should, in some way, be customizable to the main needs of the organization*"

(E14), Managers criticize the dependency on external systems and the conditional access to data, "*We are always dependent... on SPMS (the national structure that manages and develops the information systems of the Portuguese Ministry of Health) to do it*" (E9) or "*reliance on third parties (contracted companies) to obtain information that should be available*" (E13).

Perception of hospital managers on AI and its application as a support tool in decision-making processes

Participants highlighted expectations and challenges associated with the implementation, integration, and use of AI tools in decision-making in hospital management. The content analysis seems to demonstrate the potential and capacity to optimize the process and improve the efficiency of decision-making in health organizations and institutions.

Cultural expectations and human empathy

The implementation of AI solutions should be preceded by a careful analysis of real needs, as emphasized by several managers. One of them expresses the need for caution: "*I realized that it wasn't worth starting a pilot, when it is still necessary to really understand what the needs are*" (E14). This approach is crucial to ensure that innovations are relevant and effective, reflecting the operational reality of hospitals.

The long-term vision for AI is broadly positive, with managers anticipating its inevitable integration into hospital management practices: "*Understanding what is really going on, sometimes, forces us also to create tools that, perhaps in the past, we didn't even dare to think would be possible, but nowadays we can*" (E5) and "*I think in the next few years... we will have AI technologies as a decision-support system*" (E5). However, the need to culturally adapt institutions to these changes is also recognized: "*I think it is desirable and I think it is inevitable. Obviously, AI imposes some precautions since in the end the problem with AI will always be the human component.*" (E6), emphasizing that successful implementation of AI will depend on a change of mindset among managers and healthcare professionals. However, it is also clear that managers want to be part of this change: "*it seems to me that (AI) has immense potential and we cannot stop. If we stop, someone will advance sooner or later. The difference is whether we want to be dragged or be the ones dragging.*" (E15).

While operational efficiency is a clear advantage, there is substantial concern about the potential loss of "*empathy*" and human sensitivity in decisions. Managers highlight that "*to make a decision as managers, we must listen to the employees... because sometimes, what AI might be giving us, that information, can have other nuances*" (E11). This quote illustrates managers' sensitivity to the complexity of decisions in health management, which often require a deep understanding of human circumstances.

Agility and efficiency in the decision-making process

The perception of the capacity and potential of AI to streamline processes is clear and recurrent: "*It would be a valuable asset with direct application in daily decision-making.*" (E10). As described by several managers, "*We don't have the capacity to do it in a timely manner, and to do so, we have to extract several databases, cross them, to get there and, when we get to the information, the opportunity has already passed, so the time for intervention has passed and in this way, we can have it in the moment*" (E12), "*It would be much more agile for me not to lose time working on data, but to look at them and then, based on that, make real decisions*" (E1). It was also repeatedly mentioned how AI could instantly respond to operational questions, facilitating more effective and efficient management: "*Imagine: I could say 'Dr. X needs a consultation room on Monday morning. Is there a vacancy?' for example, if I could ask this question and AI answered me 'yes, from 9 am to 12:30 pm, offices Y and Z are free. Would you like to reserve one for Dr. X?' It would be fantastic*" (E1); "*Imagining a scenario in which I would have support and access to information in real-time in an integrated way... It would be fabulous.*" (E10) and "*for example, from the perspective of a central warehouse, where we can somehow anticipate that that service, on a day-to-day basis, will have a peak and articulate this with the information that it is having more complex users or is caring for more users and, therefore, this will translate into an increase in consumption.*" (E13).

AI is seen as a promising tool for optimizing human resource management. "*Establishing relationships between existing provisions, proposed by the system... revealing individual productivity by doctor*" (E2), shows how AI can facilitate more precise and well-founded management. The aggregation of these functions by AI could free managers to focus more on planning or strategy tasks, as also suggested by another participant in the study: "*such a tool would help us to really perform our role in a much more correct way*" (E14).

Several managers noted how AI could be applied: "*there are thousands of processes that need to be improved in terms of efficiency, and it is possible to take advantage of this technological side to help improve this process*" (E15). Thus: "*The areas that seem most susceptible to being used by AI algorithms in management are areas of predictive analysis*" (E5), and another adds, "*I see a lot of AI capacity used in terms of logistics principles*" (E12), "*for example, from a perspective of a central warehouse, where we can somehow anticipate that that service, on a day-to-day* *basis, will have a peak and articulate this with the information that it is having more complex users or is caring for more users and, therefore, this will translate into an increase in consumption*" (E13). These perspectives highlight the potential of AI to optimize resources and improve production.

Acess and information management

AI is perceived as a tool that can significantly simplify access to complex information, as indicated in various reviews: "*It would be a much more user-friendly scenario…*" (E2) and "*I would really like to come here to the computer and write: tell me the relationship between the prescription of medications and the comorbilities of the patients…*" (E4). These managers value the possibility of obtaining quick and accurate answers, reducing the time currently spent on complex and time-consuming analyses: "*It would ultimately facilitate my work.*" (E12) and "Move from reactive management to planned and proactive management." (E10).

A crucial point is the need for a robust data infrastructure to support AI: "*what we are saying is not science fiction… We have the data for this, and I ask these questions…*" (E4) and "*a logic where there is a single data center… That would be spectacular.*" (E7) shows the importance of a solid and well-organized database. The integration of different data sources and the robustness of the information system infrastructure are seen as essential to maximizing the potential of AI: "*AI is one of the most disruptive tools and one that will most allow us to evolve from a management perspective, both clinical and corporate, that we have ever had.*" (E13).

Integration and regulation challenges

Despite the evident enthusiasm, there is also an awareness of the challenges that AI can represent. The integration of AI is not without challenges, especially in terms of expectations versus reality: "*ChatGPT has brought the wrong thinking that a system can do everything*" (E5). Some managers even express some skepticism about the viability and effectiveness of AI: "*I use ChatGPT 4… but it has a big deficit from the point of view of what is comprehension*" (E3) and "*I confess I am a bit skeptical… I think the viability may not be what we need*" (E7). These perspectives suggest that beyond technical implementation, it is vital to consider the sustainability and real impact of AI on daily operations and the need to adjust expectations and adequately prepare the organizational culture for the changes brought by AI: "*I have no major concerns except this one, that people ask and think it is a technocratic application of AI and that can sometimes lead not to the best solutions, or that the solutions that were actually good are not implemented because they were not discussed afterwards.*" (E12). A participant states, "*Being able to anticipate what I think… I have some doubts!*" (E3), expressing caution about AI's ability to replicate complex human thought. "The gain seems to be greater than the risk." (E14) but ethical concerns are significant for most participants: "*Ethical concerns… whether in terms of data protection and privacy*" (E7), particularly the need for careful regulation of technology: "*I think it would be a valuable asset to have AI… As long as it is properly regulated*" (E2) but it's also clear that "*We should not stop innovation because of risk issues. We must analyze and understand the risks. Understand if they are acceptable within our regulatory model and essentially move forward.*" (E14).

Transformation of the managerial role

Participants anticipate a significant change in their daily roles, with AI potentially reducing the time spent on more administrative tasks, allowing greater focus on strategic activities. As mentioned by several managers, "*These days the role of the manager involves a lot of operational work*" (E13), emphasizing the time spent searching for the data and information that support the decision. "*We end up not having the time to do really what we studied for and what the institution supposedly needs from us, which is to plan, to monitor, and to improve. And we, most of the time, are not doing that*" (E1). AI is perceived as a tool that can "*help us really perform our role in a much more correct way*" (E14), optimizing planning, performance monitoring, and continuous improvement of outcomes and quality.

Some managers point to AI as a facilitator that can "*free up time to convince people, to involve people in the projects*" (E4). This perspective suggests that AI, by automating repetitive tasks, could provide more opportunities for meaningful human interactions, essential for effective change management.

Overall, the use of AI is also perceived as a support tool for the manager, but not as a replacement in the decision-making process. As one participant expressed, "*AI should always be an instrument in aiding decision-making and never the decision-making itself*" (E6) and "*In reality, decision-makers are the hardest to replace because the one thing that the machine mostly does not do is make the decision.*" (E3). This approach highlights the need to maintain human oversight over critical decisions, ensuring that technology serves as a support and not as a substitute: "*I know the business, but I know nothing about the technology and I know it exists. Someone who knows a lot about technology knows nothing about the business. And bringing these two parts together, I think we are very far from it!*" (E15).

The need for differentiated and specialized training in AI is emphasized to ensure that managers can use these technologies effectively while maintaining ethical principles: "*The main constraints result from the lack of knowledge... and with technology these issues are overcome.*" (E11). The "training for basic and advanced use of AI and digital tools" (E10) should be "*theoretical-practical*" (E2), focused not just on how to use AI, but also on understanding "*regulation and ethics in its use*" (E2). "*It is normal that people who graduated in management some time ago are not formatted to take advantage of the tools.*" (E5), and this aspect could be crucial for integrating AI responsibly into hospital management.

Maintaining the human element

Despite technological advances, it is recognized that the human and relational component in healthcare is irreplaceable: "*Healthcare today has a human and relational component that... is irreplaceable for AI*" (E6). This recognition emphasizes a balanced and cautious perspective on the implementation of AI, underlining that technology should enhance, not diminish, the human quality of hospital management: "*Our role as administrators, as managers on the boards of directors, or in another role, will continue to be very much this: that we manage, then, not to spend so much time on structuring, and that frees us up time to convince people, to involve people in what the projects we have to implement are, or the changes, or the small points*" (E4).

"*AI is one of the most disruptive tools and one that will most allow us to evolve from a management perspective, both clinical and corporate, that we have ever had*" (E13). Some managers envision a future where healthcare transcends the physical limitations of institutions, a concept described as "*the hospital without walls*" (E3). AI can support this model by facilitating quick and informed decisions, while reiterating the uniqueness and irreplaceability of human judgment: "*The worst thing we do is come up with optimal solutions for the wrong problems. And I think this is the issue: regardless of whether AI exists or not, we will still need managers, at least for the next few years. It's us (the managers) who ask the questions*." (E14).

Interviews with hospital managers about the implementation of AI in the hospital context reveal a complex panorama, illustrated in Fig. 1, where enthusiasm for technological innovation is balanced with pragmatic and ethical concerns. In terms of operational efficiency and human resource management, managers see AI as a transformative tool, capable of reducing the time spent on administrative or low-value tasks, and a way to focus on more strategic functions, such as planning and improving services: "*It will help us free up time so that we can interact more with people, because without doing that we will not be able to get the projects on the ground, get things moving.*" (E13). However, there is a cross-sectional perspective, illustrated by one of the participants: "*we end up not having the time to do really what we studied for and what the institution supposedly needs from us, which is to plan, to monitor, and to improve. And we, most of the time, are not doing that*" (E1).

On the other hand, the adequacy of the data necessary to support decision-making is a recurring concern. Participants highlight the need for robust and secure databases to maximize the effectiveness of AI integration, underscoring that any technological implementation should be preceded by a rigorous analysis of the institution's real needs. Ethical issues are also predominant, with various statements about the protection and privacy of data and the integrity of the decision occupying a central place in the different interviews. In addition, there is a clear concern about the potential loss of empathy and human contact in decisions, an aspect pointed out as fundamental in the management of organizations in general and in health institutions in particular.

Resistance to change emerged as a significant theme, with many managers expressing skepticism about the long-term viability of AI and concerns about its comprehension and interpretation capabilities in complex contexts. Adequate training in AI, covering both the use of tools and potential technological applicability as well as ethical aspects, was identified as essential for a successful integration of these tools.

The potential of AI to transform hospital management, automation, and efficiency is recognized, being pointed out as vital to ensure humanization in the decision-making process, guarantee data security, and promote a culture of acceptance of innovation. Managers unanimously consider that AI should be used as a support tool for decision-making, complementing, and not replacing, human judgment, as substantiated by the quote: "*AI should always be an instrument to assist in decision-making and never the decision-making itself*" (E6). This balance between innovation and caution/consideration is considered essential for the impact of future integration and use of AI in health organization management.

Discussion

The study explored the use of AI tools in hospital management, a field of increasing importance, with potential for change in operational efficiency and optimization of decision-making processes in health organizations and services. Across the board, the results reflect a multifaceted perspective of hospital managers on the current situation versus an innovative scenario with AI integration [32–34].

In the context of implementing AI tools in hospital management, this study made evident the high expectations of managers regarding the relative advantage of AI, somewhat aligned with perceptions revealed in the literature [32, 35–37]. Thus, although significant optimism about the potential benefits of AI is evidenced, substantial skepticism about its evidence base persists [38, 39]. The hesitation of managers to fully adopt AI, due to its early stage of implementation, reflects an important barrier that has also been observed in other research [35, 36]. This point underscores the critical need to strengthen the evidence base supporting the efficacy of AI in health management contexts [33, 40, 41].

Moreover, the notion that the complexity of implementing AI as a complement to health management is perceived as both an opportunity and a challenge is well evidenced [42]. As demonstrated, managers expressed conflicting perceptions about the complexity of AI, describing it as both inevitable and indispensable, but reiterating their low understanding and specific training on the subject. The lack of understanding about AI models emerges as a complicating element for trust in its use and full comprehension of its functionalities and practical advantages [33, 43]. This situation, while not exclusive to health institutions and managers, becomes critical due to the impact it could have on the decision to adopt it as a tool to support organizational and even clinical management, due to the indirect effects of management mechanisms on health structures [44–46]. The ethical concerns expressed by managers are evident and expected [17]. This point is fundamental not only to maintain the trust of patients and health professionals but also to ensure legal compliance, safety, and the long-term sustainability of hospital management practices supported by AI [22, 47–49].

Organizational determinants also play a crucial role. The need to deal with uncertainties related to the use and integration of AI, and the ambiguity of AI contributes to distrust among professionals [50]. Thus, it is imperative that AI implementation processes be accompanied by clear and effective change management strategies, aiming to minimize organizational resistance and maximize the acceptance of technology and innovation [35, 40, 50].

The importance of determined top leadership can significantly minimize the current challenges of the decision-making process pointed out by managers, promoting skill development and encouraging new digital transformation projects associated with management, integrating innovation such as AI [44]. Moreover, the vision of leadership is conditioned by the perception of existing technology and thus, it is imperative that healthcare organization leaders understand the capabilities, potential, and new digital trends [35]. Collaborative intelligence, which integrates human perception and analysis with AI's computational power, emerges as an essential strategy to overcome some of the barriers mentioned [46]. In this way, it will be possible to take into account the need for active involvement of managers in AI development, ensuring that technology not only supports but also amplifies analytical capacity, creating more responsive and efficient health systems [41, 46, 51, 52].

It is also noteworthy that perceptions of AI vary and are influenced by multiple factors, including personal experiences and organizational expectations [44, 53]. In this sense, future research should explore the influence of the organizational context on the adoption and use of AI, as well as defining and identifying the necessary strategies to align new technologies with the needs and expectations of various stakeholders [36, 38, 50].

Through this analysis, we aim to provide a deeper understanding of current dynamics and help shape future strategies for the adoption of emerging technologies in the health management sector [37]. The implementation of AI facilitates data-driven decision-making, offering significant potential to improve management processes in healthcare facilities, which indirectly impacts the quality of the services provided, reducing disparities [54, 55]. It is essential to overcome the identified challenges and maximize the significant benefits of using AI tools in the decision-making process in hospital management.

This study contributes significantly to the understanding of the challenges and opportunities associated with the implementation of AI tools in hospital management. By providing a detailed analysis of managers' perceptions, the research highlights key factors such as the relative advantage, complexity, and organizational determinants that influence AI adoption. The study underscores the dual nature of AI as both an opportunity and a challenge, revealing the critical need for evidence-based strategies, clear leadership, and effective change management to overcome barriers and maximize the benefits of AI integration. This research offers valuable insights that can guide future initiatives aimed at enhancing decision-making processes in healthcare organizations through AI.

Limitations

The study coincided with a period of public interest and media coverage related to AI, particularly large language models, which may have influenced the perceptions of the interviewees. This temporal context may have favored expressions of skepticism or concern about AI technology in general.

Conclusions

The use of AI tools is one of the most promising and challenging frontiers within the health sector. Through interviews with middle and senior managers, a series of insights were captured, reflecting the current decision-making process, its challenges and limitations, and the perception of potential for change and improvement with the use of AI tools. The integration of AI in hospital management is perceived by managers as a necessary and inevitable evolution that will bring various benefits in terms of efficiency, accessibility of information, and analytical capacity. However, to fully realize these benefits, it is essential that institutions prepare an adequate foundation, both in terms of data infrastructure and the training and cultural adaptation of their professional managers. This preparation will help maximize the benefits and minimize the challenges associated with this technological and organizational transition.

This study underscores the complexity and dynamism of AI integration, highlighting a future where technology and the humanization of hospital management should advance hand in hand. To achieve this, healthcare organizations should invest in robust data systems and ensure interoperability to provide managers with seamless access to accurate and real-time information. Additionally, targeted training programs are critical to equip managers with the skills necessary to effectively use AI tools, ensuring that these technologies enhance rather than replace human judgment.

Moreover, fostering a culture of collaboration and open communication among managers can address the current isolation in decision-making processes, allowing for more integrated and well-founded decisions. This study provides a fundamental step towards understanding the transformative potential of AI in the current decision-making process, particularly in relation to the expectations and difficulties pointed out by managers. It also explores how to ensure the ethical, safe, efficient, and effective integration of AI as a decision-support tool in managing health institutions.

By addressing these areas, healthcare managers can enhance decision-making processes, improve operational efficiency, and optimize resource allocation, ultimately contributing to more effective and humane healthcare delivery. The findings of this study offer actionable insights that can guide stakeholders in navigating the complexities of AI adoption in hospital management, ensuring that technology serves as a valuable complement to human expertise.

Abbreviations
AI Artificial Intelligence

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Additional file 1. Interview guide.

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Authors' contributions
MA conducted data collection and composed the manuscript; JS and TM edited and critically reviewed the manuscript and provided guidance; TS provided guidance. All authors read and approved the final manuscript.

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Availability of data and materials
Empirical material generated and/or analyzed during the current study are not publicly available, but are available from the corresponding author on reasonable request.

Ethics approval and consent to participate
The study was based on original data. The authors confirm that all methods were carried out in accordance with relevant guidelines and regulations and confirm that informed consent was obtained from all participants. Ethics approval was granted by the Ethics Committee of NOVA-NSPH, Lisbon, Portugal (ID:09/2024).

Consent for publication
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Competing interests
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